



Quiz : Annexure A MCQ Test
Date : Wed Aug 12 2026 1:23 PM
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Accuracy

97%

Total Questions

30

✓ Correct

29

× Incorrect

1

No.	Question	Time	Points	Response
1	Consider a simple linear regression model with one independent variable. Which of the following is true?	14	1	The regression line ✓ (a) always passes through the mean of X and Y.
2	Which of the following statements is correct about Decision Trees?	21	1	Decision trees can be ✓ (c) used for classification and regression.
3	Which of the following methods can be used for pruning in decision trees?	16	1	✓ (c) Both (a) and (b)
4	Which of the following is a disadvantage of Decision Trees?	16	1	✓ (a) Prone to overfitting.
5	What does Bayes' Theorem primarily help with in probability and statistics?	14	1	✓ (b) Updating probabilities based on new evidence
6	In Bayes' Theorem, what does the term $P(A B)$ represent?	16	1	The probability of event ✓ (b) A given event B has occurred
7	A factory has two machines, A and B. Machine A produces 60% of the products, and machine B produces 40%. If 2% of A's products and 3% of B's products are defective, what is the probability that a randomly selected defective product came from Machine B?	38	1	✓ (b) 0.50
8	Which of the following is not required to apply Bayes' Theorem?	19	1	✓ (c) Total number of possible outcomes
9	A doctor uses a test to diagnose a disease that affects 1% of a population. The test correctly identifies the disease 95% of the time but also gives a false positive in 5% of cases. If a patient tests positive, what is the probability they actually have the disease?	14	1	✓ (a) 0.16
10	Bayes' Theorem is most useful when dealing with which type of probability?	15	1	✓ (b) Conditional probability
11	If a hypothesis HHH has a prior probability of 0.3 and new evidence EEE has a likelihood of 0.8 given HHH, how does Bayes' Theorem adjust the probability of HHH after seeing EEE?	3	0	✗ (d) The probability of HHH is completely unknown
12	A bag contains 5 red and 7 blue balls. A ball is drawn randomly and found to be red. What is the probability that the next ball drawn is also red, assuming the first ball is not replaced?	16	1	✓ (b) $\frac{4}{11}$

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13	Which field heavily relies on Bayes' Theorem for updating beliefs based on new data?	17	1	✓ (a) Machine Learning
14	Why is backpropagation crucial in training deep neural networks?	16	1	It updates weights using ✓ (b) the gradient of the loss function.
15	During backpropagation, what happens when the learning rate is too high?	24	1	The model may overshoot the optimal weight values and fail to converge. ✓ (b)
16	What is the main advantage of rule-based induction models?	14	1	✓ (d) Both (a) and (b)
17	Which of the following is an example of a rule-based induction algorithm?	14	1	✓ (a) Decision Tree
18	In rule-based induction, what happens when the rules are too specific to the training data?	15	1	✓ (a) The model may suffer from overfitting.
19	Which approach is commonly used to simplify rule-based induction models?	15	1	✓ (a) Pruning redundant or less significant rules
20	What is the primary purpose of a confusion matrix in evaluating a classifier?	15	1	To visualize the performance of a classifier by showing correct and incorrect predictions ✓ (a)
21	Which metric is used to measure the proportion of correctly classified instances out of the total instances?	14	1	✓ (c) Accuracy
22	In a binary classification problem, what does the True Positive (TP) value represent?	16	1	The number of positive instances correctly classified as positive ✓ (b)
23	What is the formula for calculating precision in a classification problem?	11	1	Precision = $\frac{TP}{TP + FN}$ ✓ (a)
24	Which of the following metrics is most useful when the classes are imbalanced?	21	1	✓ (b) F1-Score
25	What does a high recall value indicate in a classification model?	16	1	The model has a low number of false negatives ✓ (b)

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26	Which of the following is an example of a rule-based induction algorithm? (CO3)	15	1	✓ (a) Decision Tree
27	In rule-based induction, what happens when the rules are too specific to the training data? (CO3)	14	1	✓ (a) The model may suffer from overfitting.
28	Which approach is commonly used to simplify rule-based induction models? (CO3)	16	1	✓ (a) Pruning redundant or less significant rules
29	What is the primary purpose of a confusion matrix in evaluating a classifier? (CO3)	15	1	✓ (a) To visualize the performance of a classifier by showing correct and incorrect predictions
30	Which metric is used to measure the proportion of correctly classified instances out of the total instances? (CO3)	14	1	✓ (c) Accuracy